ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q3, Q4

Question 1

(a) (i) $f_x = f / p = 16\ \text{mm} / 5\ \mu\text{m} =$ **3200 px**.

(ii) $W_s = 1280 \times 5\ \mu\text{m} = 6.4$ mm. FOV $= W_s \cdot Z/f = 6.4 \times 400/16 =$ **160 mm**.

Per pixel $= 160/1280 \approx$ **0.13 mm/px**.

(b) LSB $= 5/2^{12} = 5/4096 = 1.2207$ mV.

$\Delta F = \text{LSB}/S = 1.2207/10 =$ **0.122**

(c) ZOH discretisation: $F = e^{AT}$, $G = \int_0^T e^{A\tau} B\, d\tau = A^{-1}(F - I)B$ $(A \neq 0)$.

Here $A = -2$, $B = 1$, $T = 0.1$: $F = e^{-0.2} =$ **0.8187**. $G = (F - 1)/A = (0.8187 - 1)/(-2) =$ **0.0906**.

$x_{k+1} = 0.8187\, x_k + 0.0906\, u_k$.

Question 3

(a) (1) Differentiation is a high-pass operation (gain $\propto \omega$), so encoder measurement/quantisation noise is amplifed, by about $1/T = 1000$ at 1 kHz.

(2) Encoder quantisaton: $\Delta\theta$ is a whole number of counts, so $\hat{\omega} = \Delta\theta/T$ moves in steps of (count size)/T; at low speed the estimate is a coarse staircase of zeros and spikes.

(b) $\omega_c = 20$ rad/s. Tustin with $T = 0.001$: $a = \omega_c T/2 = 0.01$.

$y_k = [a/(1 + a)](u_k + u_{k-1}) + [(1 - a)/(1 + a)]\, y_{k-1} =$ **$0.00990\,(u_k + u_{k-1}) + 0.98020\, y_{k-1}$**.

DC gain: $2(0.0099)/(1 - 0.9802) = 0.0198/0.0198 = 1$ ✓.

(c) $A - LC = [-\ell_1, 1; -\ell_2, -2]$. $\det(sI - A + LC) = (s + \ell_1)(s + 2) - \ell_2 = s^2 + (2 + \ell_1)s + (2\ell_1 - \ell_2)$.

Desired $(s + 10)^2 = s^2 + 20s + 100$. So $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$ and $2\ell_1 - \ell_2 = 100 \Rightarrow \ell_2 = 36 - 100 = -64$.

$L = [18, -64]^T$.

(d) Advantage of the observer: it uses the model and the input u, so there is no lag in the estimate, which a filter always introduces.

The fliter is safer when the sampling rate is very high, because the observer would not be able to run fast enough.

Question 4

(a) Closed loop: $\ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\, r$, i.e. $s^2 + 4s + 8 = 0$ so the poles are $s = -2 \pm 2j$.

DC gain $= bN_{ff}/(c + bk_1) = N_{ff}/8$. For unity gain $N_{ff} = 1/8 =$ **0.125**.

(b) $G_{DC} = bN_{ff}/(c + bk_1)$. Numerator: $0.75 \times 8 = 6$. Denominator: $c + k_1 = 8$ as before.

$\theta_{ss} = 6/8 =$ **0.75**, so error = **25%**.

(c) Char. poly: $s^3 + (a + bk_2)s^2 + (c + bk_1)s + bK_I$.

Desiered: $(s + 4)(s^2 + 4s + 8) = s^3 + 4s^2 + 8s + 4s^2 + 16s + 32 = s^3 + 8s^2 + 16s + 32$ (combining terms).

$2 + k_2 = 8 \Rightarrow$ **$k_2 = 6$**; $6 + k_1 = 16 \Rightarrow$ **$k_1 = 10$**; $K_I =$ **32**.

(d) The integrator keeps accumulating the error until the error becomes zero, so the stedy-state error is zero whatever b and c are, if the loop is stable.

Failure mode: integrator windup when the actuator saturates. Remidy: anti-windup, e.g. clamp the integrator.